PI7C9X3G606GP / PI7C9X3G808GP / PI7C9X3G816GP / PI7C9X3G1632GP
Diodes Gen3 PCIe® Packet Switch for Up Port Selection Application

Introduction

Diodes Gen3 PCIe Packet Switch supports Up Port Selection function. This function allows any port to be an upstream port in Fan-out Mode.

Up-port Select functionality determined by Reg 504h bit[2:0] (Port 0 Only):

9.3.186 DEVICE CONFIGURATION 0 REGISTER – OFFSET 504h (Port 0 Only)

BIT	FUNCTION	TYPE	DESCRIPTION	EEPROM/ I2C-SMBUS	DEFAULT
2:0	Up Port Selection	RO	Used to do up port selection. It is valid for transparent mode only.	Yes	000b

To realize Up port Select function, there are two ways：

a) By EEPROM
b) By I2C command

The following section will briefly describe how to implement the Up-port Select function in the above two ways.

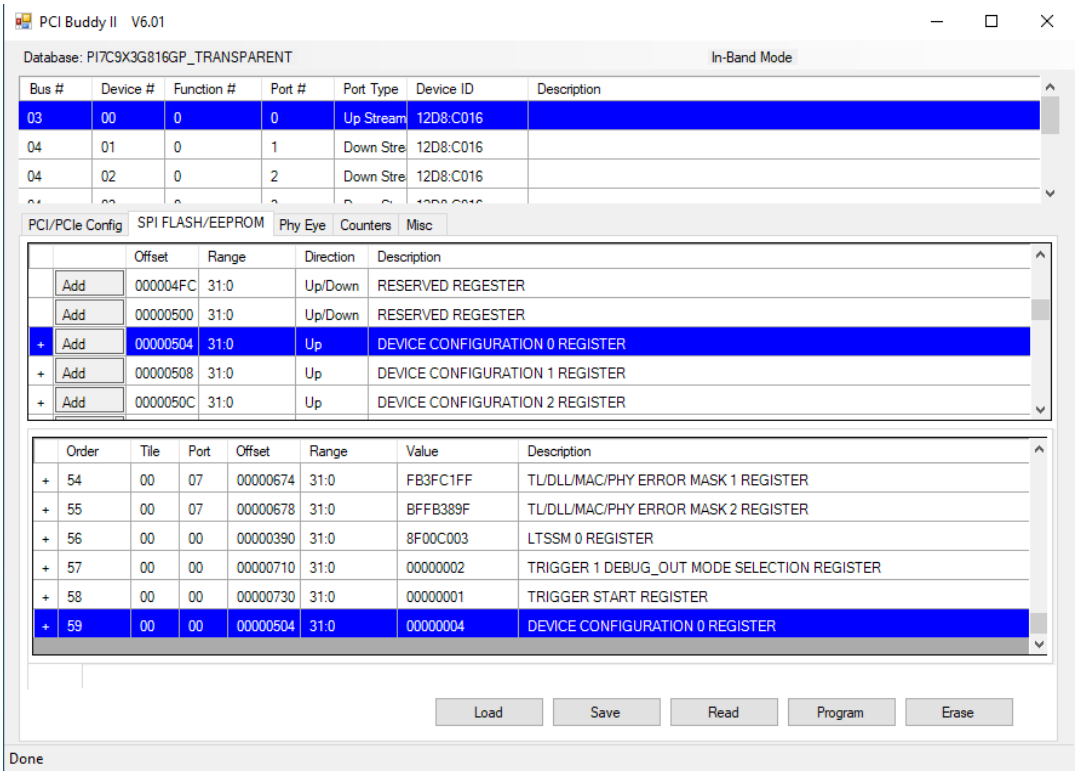
By EEPROM:

To realize the Up-Port Select function through EEPROM, the EEPROM content can be modified by PCIBUDDY, which is a tool developed by Diode for Diodes Gen3 Packet Switch.

Steps:

a) Plug in DUT with Diodes Gen3 Packet Switch in Platform
b) Open PCIBUDDY (Note 1), and click “SPI FLASH/EEPROM” sheet
c) Find 504h in the middle block, and press “Add” to add Reg 504h into the bottom block (EEPROM content)
d) Modified 504h bit[2:0] to the corresponding port which you want to be the Up-port (Make sure to modify 504h at Port 0)
e) Re-boot system

Following is example for 64-bit Windows system with In-Band Mode.



Note: For In-Band Mode, PCIBUDDY only support 64-bit Win OS. If your system is 32-bit Win OS or non-Windows system, please open PCIBUDDY via I2C for Out-Band Mode. Detailed information please refer to PCIBUDDY User Manual

By I2C:

To realize the Up-Port Select function through I2C, I2C commands are as following：(Use 3G816 for example, change up-port from P0 to P4, and the from P4 to P0)

Steps:

Up-port Select change by I2C from P0 to P4

1) Write LINK CONTROL register2 (98h bit[3:0]=1b) at port 0: set Port 0 target link speed to GEN1
2) Write LINK CONTROL register (78h bit[5]=1b) at port 0: retrain the link
3) Write LTSSM 0 register (390h bit[2]=1b) at port 0: force Port 0 LTSSM state stay in detect state
4) Write LINK CONTROL register2 (98h bit[3:0]=1b) at port 4: set Port 4 target link speed to Gen1
5) Write LINK CONTROL register (78h bit[5]=1b) at port 4: retrain the link
6) Write LTSSM 0 register (390h bit[2]=1b) at port 4: force Port 4 LTSSM state stay in detect state
7) Write CFG0 register (504h bit[2:0]=100b) at port 0: select Port 4 as upstream port
8) Write PCIE CAP register (68h bit[23:20]=5, bit[24]=0b) at port 4: to set port/device type/function to PCIe switch upstream port
9) Write PCIE CAP register (68h bit[23:20]=6, bit[24]=1b) at port 0: to set port/device type/function to PCIe switch downstream port
10) Write LINK CONTROL register2 (98h bit[3:0]=3b) at port 0: set Port 0 target link speed to GEN3
11) Write LTSSM 0 register (390h bit[2]=0b) at port 0: disable forcing Port 0 LTSSM state stay in detect state
12) Write LINK CONTROL register2 (98h bit[3:0]=3b) at port 4: set Port 4 target link speed to GEN3
13) Write LTSSM 0 register (390h bit[2]=0b) at port 4: disable forcing Port 4 LTSSM state stay in detect state
14) Change RC 18h：18h needs to re-setting as Port 4 becomes to upstream port
15) Write BUS Number register (18h) of Port 4 as Port 4 becomes to upstream port

Up-port Select change by I2C from P4 to P0

1) Write LINK CONTROL register2 (98h bit[3:0]=1b) at port 0: set Port 0 target link speed to GEN1
2) Write LINK CONTROL register (78h bit[5]=1b) at port 0: retrain the link
3) Write LTSSM 0 register (390h bit[2]=1b) at port 0: force Port 0 LTSSM state stay in detect state
4) Write CFG0 register (504h bit[2:0]=000b) at port 0: select Port 0 as upstream port
5) Write LINK CONTROL register2 (98h bit[3:0]=1b) at port 4: set Port 4 target link speed to Gen1
6) Write LINK CONTROL register (78h bit[5]=1b) at port 04: retrain the link
7) Write LTSSM 0 register (390h bit[2]=1b) at port 4: force Port 4 LTSSM state stay in detect state
8) Write PCIE CAP register (68h bit[23:20]=5, bit[24]=0b) at port 0: to set port/device type/function to PCIe switch upstream port
9) Write PCIE CAP register (68h bit[23:20]=0, bit[24]=0b) at port 4: to set port/device type/function to PCIe switch downstream port
10) Write LINK CONTROL register2 (98h bit[3:0]=3b) at port 4: set Port 4 target link speed to GEN3
11) Write LTSSM 0 register (390h bit[2]=0b) at port 4: disable forcing Port 4 LTSSM state stay in detect state
12) Write LINK CONTROL register2 (98h bit[3:0]=3b) at port 0: set Port 0 target link speed to GEN3
13) Write LTSSM 0 register (390h bit[2]=0b) at port 0: disable forcing Port 0 LTSSM state stay in detect state

14) Set RC 18h: 18h needs to re-setting as Port 0 becomes to upstream port
15) Write BUS Number register (18h) of Port 0 as Port 0 becomes to upstream port

Note1: Up Port Selection for PI7C9X3G1632GP (two tiles) Supplementary

For PI7C9X3G1632GP, if up port is not at T0_P0, except offset 504h, it also need set offset 558h for virtual up/dn mapping. Below is an example if up port is changed to T1_P0.

1) Write DEVICE CONFIGURATION 0 register, offset 504h
   bit[5:0]: up port selection, for Tile 0 and Tile 1, P0 (Tile0 = Tile1 setting)

2) Write SWITCH DOMAIN MODE CONTROL register, offset 558h
   bit[6:0]: enable virtual Up/Dn
   - If up is set to T0_P0
     T0_P0 (master), offset 558h, bit[0]=0, bit[1]=1
     T1_P0 (slave), offset 558h, bit[0]=1, bit[6:1]=0
   - If up is set to T1_P0
     T0_P0 (slave), offset 558h, bit[0]=1, bit[6:1]=0
     T1_P0 (master), offset 558h, bit[0]=0, bit[1]=1